CHAPTER 8

USING MACROS TO EDIT

In this chapter you'll learn about

- Recording a Macro
- Playing a Macro
- Editing a Macro
- Creating a Macro

Macros help you automate word processing tasks. They can be used to type frequently used text or to execute a sequence of commands. You create a macro by recording and saving your keystrokes. When you play the macro, the keystrokes that you recorded are carried out. You can edit the macros to correct any errors or customize them.

RECORDING A MACRO

When you record a macro, WordStar captures and saves the keystrokes in a macro file. When you press the hot key you assign to the macro, the macro executes the keystrokes that you saved.

Follow the steps below to record a macro that saves the current file, opens a window on another file, and sizes the window. You'll call the macro "Window."

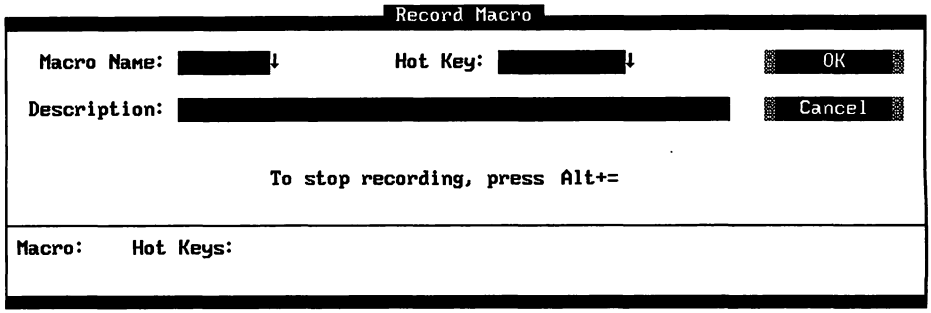
1 Choose File→Open Document (F,D) and name the file MACRO.DOC.

2 Type the sample text below:

```
WordStar stores each macro as a file, with the name you
gave it while recording or editing and the .WSM extension.
```

3 Choose Utilities→Macros→Record (Alt+U,M,R) to record a macro.

The Record Macro dialog box appears.



4 In the Macro Name box, type **window**.

Note: When you name a macro, use a name that will help you identify it. You can use up to eight characters. WordStar assigns the .WSM extension to the filename.

5 Press the Tab key.

The cursor moves to the Hot Key box. Now you'll assign the single key or combination of keys for running the macro. For more information about hot key assignments, see the *Reference* manual.

6 Press the Down Arrow to choose a hot key from the list of available hot keys.

7 In the Hot Key list, move the cursor to Alt+W and press Enter.

Alt+W appears in the Hot Key box and the cursor moves to the Description box.

8 In the Description box, type a description of the macro. Type **Save file-Open window-Size window.**

You can use as many as 48 characters, including spaces, for a macro description.

Now that you've named and described the macro, you're ready to record the keystrokes.

9 Press F10 or choose OK to begin recording keystrokes.

(Ignore the *End Recording* message at the top of the screen.)

Now you'll record the following sequence of keystrokes for the macro:

1 Press Alt+F,S to save the file MACRO.DOC.

2 Press Alt+F,W to open a window on a second file.

The Open Document dialog box appears.

3 Press Alt+Hyphen (hold down the Alt key while you press the hyphen key) to insert a pause for input command into the macro. When you play back the macro later, the macro will pause while you enter a filename.

4 Type **REVIEW.DOC** for the second filename.

5 Press Alt+hyphen to end the pause for input.

The macro resumes recording.

6 Press F10 or choose OK to open REVIEW.DOC.

7 Press Alt+V,W to size the window.

8 In the Change Window Size dialog box, press F to make each window use the full screen. Press F10 or choose OK to close the dialog box.

Each window now uses the full screen.

9 Press Alt+Equal (hold down the Alt key while you press the Equal key) to stop recording.

10 Close and abandon REVIEW.DOC. Then close and abandon MACRO.DOC.

You now have a macro that saves a file, opens a second window, and sizes the window to fill the entire screen.

PLAYING A MACRO

When you *play* a macro, the commands recorded in the macro are carried out. You can play a macro by using a menu command or the assigned hot key. In the following section, you'll play the WINDOW macro by using the hot key you assigned.

1 Open the MACRO.DOC file.

2 Press Alt+W to play the WINDOW macro. When the Open Document dialog box appears, the macro pauses for input.

3 Type a filename for the second window and press Alt+Hyphen to end the pause for input. The second window is opened and sized.

4 Close and abandon both files.

EDITING A MACRO

After you record a macro, you may want to change it. WordStar has a macro language that you can use to edit a macro.

Follow these steps to edit a macro:

1 At the Opening screen, choose Utilities→Macros→Edit/Create (U,M,E).

The Edit/Create Macro dialog box appears.

2 In the Macro Name box, type **window**.

3 Press F10 or choose OK to edit the macro.

The WINDOW macro (WINDOW.WSM) is opened.

In every macro file, you'll see the command "Sub MAIN" which is a subroutine. A subroutine is a collection of statements that are invoked by the Sub command. Each macro must contain the Sub MAIN subroutine.

Subroutines usually contain key functions. A key function tells the macro which keys to play back. The format for a key function is:

```
Key("key stroke string")
```

In the Window macro, you'll see these key functions near the end of the file.

```
KEY ("{F10}{Alt+V}w")
KEY ("f")
```

The first key function above displays the Change Window Size dialog box. The second key function changes the window size to full screen.

In the following steps, you'll put a comment at the beginning of the WINDOW.WSM file to provide descriptive information about the macro. A comment is a statement for you to read but is not a command for the macro to execute.

1 At the beginning of line 1, type this text:

```
rem  Saves current file, opens and sizes second window.
```

2 Press Enter.

In the following steps, you'll change a key function so that the window takes up only five lines instead of the full screen.

The key function looks like this:

```
KEY ("f")
```

1. Move the cursor to the "f" in the key function.
2. Delete the "f" and type **L5**.
3. Choose File→Save and Close (Alt+F,L) to save the macro.
4. Open a file and the WINDOW macro to see the changes you just made. (If you need help, see "Playing a Macro" in this chapter.)

   If WordStar finds errors, it returns you to the macro file to correct them.

CREATING A MACRO WITH THE MACRO LANGUAGE

In this chapter, you've created a macro by recording your keystrokes. Now you'll create a macro using the macro language.

In the following steps, you'll create a macro that runs a spelling check, saves your document, and then runs Preview.

1. At the Opening screen, choose Utilities→Macros→Edit/Create (U,M,E).
2. Assign a macro name, for example, SPELL. WordStar gives the file a .WSM extension.
3. Assign a hot key such as Alt+3 and type a description.
4. Press F10 or choose OK to open the macro.

   WordStar opens the new macro.
5. On line 1, type this text:

```
rem  Runs spelling check, saves file, runs Preview.
```

6. Press Enter.

7 Type **Sub MAIN** and press Enter. This marks the beginning of the main subroutine.

8 On the next two lines, type the sample statements shown below. (You can type these statements in all lowercase; the case used below is for readability only.)

Note: "AutoRestore(ON)" is an example of a statement. When there is more than one statement on a line, the statements must be separated by colons. When you type statements, separate them with a space, a colon, and another space.

```
AutoRestore(ON) : SetHelpLevel(4) : Insert(ON)
CmdTags(OFF) : HideDots(ON) : ColMode(OFF) :
ColReplace(OFF)
```

9 Press Enter.

The statements establish these settings:

AutoRestore(ON) Restores original settings after the macro plays.

SetHelpLevel(4) Sets the help level at 4.

Insert(ON) Allows characters to be inserted into existing text instead of overwriting text.

CmdTags(OFF) Turns off the display of command tags.

HideDots(ON) Turns off the display of dot commands.

ColMode(OFF) Turns column mode off. This is the default for marking blocks.

ColReplace(OFF) Turns off column replace. This is the default for moving and copying blocks.

In the following steps, you'll type two key functions and some exception conditions:

The first key function that you'll type is:

```
Key("{Alt+U}S")
```

1 Type **Key("** to begin the Key function. Every Key function begins with **("** and ends with **")**.

2 Type **{Alt+U}S** to move the cursor to the beginning of the file and start the spelling check.

3 Type **")** to end the function. Press Enter to move to the next line. Press the Tab key to indent the line.

4 Type **IfException** and press Enter.

When a spelling check is complete, WordStar displays a message giving the word count. This is an exception condition. When a macro encounters an exception condition, you can have the macro pause while you provide required input or you can tell the macro to provide the input. To have the macro provide input, you add an IfException clause to your macro statement. In the following steps, you tell the macro to issue an Escape when the word count message is displayed.

5 On the next line under "IfException" type **ACK: Key("{Esc}")** and press Enter.

The ACK clause issues an Escape key command when the spelling check message appears.

6 To end the IfException clause type **End IfException** and press Enter.

7 Type **Key("{Alt+F}S")** and press Enter. This key function saves the file.

8 Type **Key("{Alt+V}P")** and press Enter. This key function runs Preview.

9 Type **End Sub** to end the subroutine and press Enter.

When your macro is finished, it will look like this:

```
rem  Runs spelling check, saves file, runs Preview
Sub MAIN
AutoRestore(ON) : SetHelpLevel(4) : Insert(ON) :
CmdTags(OFF)
HideDots(ON) : ColMode(OFF) : ColReplace(OFF)
Key("{Alt+U}S")
          IfException
          ACK: Key("{Esc}")
End IfException
Key("{Alt+F}S")
Key("{Alt+V}P")
End Sub
```

10 Save the macro and then play it in a document.